

# Asymptotic Inversion of the Gamma and Barnes $G$ Functions

## Lambert-Normalized Transseries, Explicit Coefficients, and a General Power–Logarithmic Reversion Calculus

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### Abstract

The large-value inverses of the Gamma function and the Barnes  $G$ -function are governed by a common nonlinear scale: after taking a logarithm, their dominant terms are respectively  $x \log x$  and  $\frac{1}{2}x^2 \log x$ . We develop a systematic inversion calculus for asymptotic transseries whose leading block is

$$ax^P(\log x - \beta),$$

or, more generally,  $ax^P(\log x - \beta)^P$ . The dominant block is inverted exactly by a suitable branch of Lambert's  $W$ -function. If  $X$  denotes this exact leading inverse and  $q = X^{-1}$ , the remaining inverse is a triangular series

$$x = X + \sum_{n \geq 0} C_n q^n$$

with coefficients in a slow differential field generated by the Lambert coordinate. We prove formal existence and uniqueness, give a coefficient recurrence and its first universal terms, derive a near-identity Lagrange–Buermann representation, describe Newton acceleration and residual certification, and explain how algebraic and exponentially small sectors are transported under inversion.

For the increasing real Gamma branch, with

$$T = \log\left(\frac{y}{\sqrt{2\pi}}\right), \quad w = W_0(T/e), \quad X = T/w, \quad s = w + 1,$$

we obtain

$$\Gamma_+^{(-1)}(y) = \frac{1}{2} + X + \frac{1}{24sX} - \frac{14s^2 + 10s + 5}{5760s^3 X^3} + \dots,$$

and display the expansion through  $X^{-9}$ . Centering the Stirling series at  $x - \frac{1}{2}$  forces every even inverse-power coefficient to vanish. We also give convergent large- $y$  expansions for all inverse branches tending to poles of  $\Gamma$ , including the branch tending to zero.

For the eventual positive Barnes branch, with

$$Y = \log y, \quad w = W_0(4Ye^{-3}), \quad Z = 2\sqrt{Y/w}, \quad D = w + 1,$$

we derive

$$G_+^{(-1)}(y) = 1 + Z - \frac{\log(2\pi)}{D} + \frac{C_1(D)}{Z} + \frac{C_2(D)}{Z^2} + \frac{C_3(D)}{Z^3} + \dots,$$

with explicit rational functions  $C_1, C_2, C_3, C_4$ . Re-expansion of the Lambert coordinate gives conventional nested-logarithm transseries, while retaining  $W$  exactly preserves the correct sector ordering. Exact symbolic substitution and high-precision numerical tests verify all reported coefficients.

**Principal real branches at a glance**

Write compositional inverses with the symbol  $f^{(-1)}$ , so that they cannot be confused with reciprocals. For  $y \rightarrow +\infty$ , the increasing real Gamma branch is

$$\begin{aligned} \Gamma_+^{(-1)}(y) = & \frac{1}{2} + X + \frac{1}{24sX} - \frac{14s^2 + 10s + 5}{5760s^3X^3} \\ & + \frac{2232s^4 + 1176s^3 + 714s^2 + 350s + 105}{2903040s^5X^5} \\ & - \frac{822960s^6 + 292536s^5 + 136956s^4 + 68040s^3 + 32970s^2 + 12250s + 2625}{1393459200s^7X^7} \\ & + \frac{309052800s^8 + 77920128s^7 + 26024592s^6 + 10019592s^5 + 4516028s^4}{367873228800s^9X^9} \\ & + \frac{2023560s^3 + 812350s^2 + 242550s + 40425}{367873228800s^9X^9} + O\left(\frac{1}{sX^{11}}\right), \end{aligned}$$

where

$$T = \log(y/\sqrt{2\pi}), \quad w = W_0(T/e), \quad X = T/w, \quad s = w + 1.$$

For the eventual positive Barnes branch, set

$$Y = \log y, \quad w = W_0(4Ye^{-3}), \quad Z = 2\sqrt{Y/w}, \quad D = w + 1,$$

and put  $\lambda = \log(2\pi)$ ,  $\alpha = \log A$ , where  $A$  is Glaisher's constant. Then

$$G_+^{(-1)}(y) = 1 + Z + C_0 + \frac{C_1}{Z} + \frac{C_2}{Z^2} + \frac{C_3}{Z^3} + O(Z^{-4}),$$

with

$$\begin{aligned} C_0 &= -\frac{\lambda}{D}, \\ C_1 &= \frac{1}{12} + \frac{2\alpha}{D} + \frac{\lambda^2}{2D^2} - \frac{\lambda^2}{D^3}, \\ C_2 &= \frac{4\lambda\alpha}{D^3} + \frac{4\lambda^3}{3D^4} - \frac{2\lambda^3}{D^5}, \end{aligned}$$

and  $C_3$  is displayed in [Theorem 8.1](#).

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# 1 Introduction

The Gamma and Barnes  $G$ -functions grow faster than exponentials but more slowly than iterated exponentials. Their logarithms have the leading forms

$$\log \Gamma(x) = x \log x - x + \text{lower blocks}, \quad (1.1)$$

$$\log G(x+1) = \frac{1}{2}x^2 \log x - \frac{3}{4}x^2 + \text{lower blocks}. \quad (1.2)$$

Consequently, their large-value inverse functions are not described by a plain Puiseux or inverse-power series. The first inversion already requires Lambert's  $W$ -function; after this exact normalization, the remaining corrections form a power series in the reciprocal leading inverse, with coefficients that vary only logarithmically.

The leading Lambert approximation to the principal inverse Gamma function is classical and is discussed, for example, by Borwein and Corless and by Jeffrey and Watt [2, 10]. The analytic branch structure of inverse Gamma functions was studied by Uchiyama and Pedersen [16, 14]. Our purpose here is different: we carry the large-argument inversion to arbitrary asymptotic order, make the coefficient recursion explicit, apply the same machinery to Barnes  $G$ , and isolate a reusable theory for power-logarithmic transseries.

The linked ProveIt drafts develop a broad calculus for logarithmic transseries and near-identity series reversal [15]. The present article uses the same organizing principle but adapts the coordinate system to maps whose leading derivative is not asymptotic to one. The decisive step is to invert the dominant power-logarithmic block *exactly*. In the resulting Lambert coordinate, the correction problem becomes near-identity and triangular.

## 1.1 What is meant by a transseries here

Three nested levels occur.

1. The *Lambert-normalized outer expansion* is a series in  $q = X^{-1}$ , where  $X$  is the exact inverse of the dominant block. Its coefficients are rational functions, or more general slow transseries, in a Lambert coordinate  $w \sim \log Y$ .
2. Expanding  $W$  itself in  $\log Y$  and  $\log \log Y$  produces a conventional logarithmic transseries.
3. The Stirling series are factorially divergent. Exponential improvement adds beyond-all-orders sectors; inversion transports each such sector by division by the derivative of the algebraic part, followed by triangular nonlinear coupling.

The first level is normally the best representation for computation and for clean asymptotic bookkeeping. The second is useful when an answer is required entirely in elementary iterated logarithms. The third records the optimal-truncation and Stokes structure.

## 1.2 Branch conventions

On the positive real axis,  $\Gamma$  decreases from  $+\infty$  at  $0^+$  to its unique positive minimum and then increases to  $+\infty$ . Thus a sufficiently large positive value has two positive preimages. We write

- $\Gamma_-^{(-1)}(y) \rightarrow 0^+$  for the left branch;
- $\Gamma_+^{(-1)}(y) \rightarrow +\infty$  for the increasing, or principal, branch.

In addition, there are inverse branches approaching every pole  $-n$ ,  $n = 0, 1, 2, \dots$ . These pole branches have convergent expansions in  $1/y$ , unlike the Stirling branch at  $+\infty$ .

The Barnes  $G$ -function is not globally one-to-one on the positive axis, but  $(\log G)'(x) \sim x \log x$ ; hence it is strictly increasing for all sufficiently large  $x$ . We denote its eventual positive inverse by  $G_+^{\langle -1 \rangle}$ .

For complex inversion, one first chooses a branch of  $\text{Log } y$ , then a branch  $W_k$ , and finally a sector in which the chosen Stirling expansion and logarithm are consistent. The coefficient formulae below are unchanged; only the slow coordinates and branch labels change.

### 1.3 Summary of the method

The reusable pipeline is:

1. take a logarithm of the target special function when appropriate;
2. translate the argument and subtract constants to put the forward expansion in a sparse normal form;
3. invert the leading block by Lambert  $W$ ;
4. set the true inverse equal to the leading inverse plus an additive correction;
5. expand the normalized residual in  $q = X^{-1}$  and solve one linear equation per coefficient;
6. certify the truncation by substitution or by one Newton step.

## 2 Forward asymptotics and normal forms

### 2.1 Gamma in the half-shifted coordinate

The standard Stirling expansion is [6, 12]

$$\log \Gamma(x) \sim \left(x - \frac{1}{2}\right) \log x - x + \frac{1}{2} \log(2\pi) + \sum_{k \geq 1} \frac{B_{2k}}{2k(2k-1)x^{2k-1}}. \quad (2.1)$$

For inversion, the translated coordinate

$$z = x - \frac{1}{2} \quad (2.2)$$

is substantially better. The Bernoulli-polynomial form of Stirling's series gives

$$\log \Gamma\left(z + \frac{1}{2}\right) \sim z(\log z - 1) + c_\Gamma + \sum_{k \geq 1} \frac{a_k}{z^{2k-1}}, \quad c_\Gamma = \frac{1}{2} \log(2\pi), \quad (2.3)$$

where

$$a_k = \frac{B_{2k}(1/2)}{2k(2k-1)} = \frac{(2^{1-2k} - 1)B_{2k}}{2k(2k-1)}. \quad (2.4)$$

The first values are

$$a_1 = -\frac{1}{24}, \quad a_2 = \frac{7}{2880}, \quad a_3 = -\frac{31}{40320}, \quad a_4 = \frac{127}{215040}, \quad a_5 = -\frac{511}{608256}. \quad (2.5)$$

Only odd inverse powers occur. This sparsity propagates to the inverse series and is the main reason for centering at one half.

### 2.2 Barnes $G$ in the natural shifted coordinate

Barnes' function satisfies

$$G(z+1) = \Gamma(z)G(z), \quad G(1) = 1. \quad (2.6)$$

For  $|\arg z| \leq \pi - \delta$ , its logarithm has the asymptotic expansion [7, 11]

$$\log G(z+1) \sim \frac{1}{2}z^2 \log z - \frac{3}{4}z^2 + \frac{\lambda}{2}z - \frac{1}{12} \log z + \kappa + \sum_{k \geq 1} \frac{b_k}{z^{2k}}, \quad (2.7)$$

$$\lambda = \log(2\pi), \quad \kappa = \zeta'(-1) = \frac{1}{12} - \alpha, \quad \alpha = \log A, \quad (2.8)$$

$$b_k = \frac{B_{2k+2}}{2k(2k+2)}. \quad (2.9)$$

Here  $A$  is Glaisher's constant. Thus

$$b_1 = -\frac{1}{240}, \quad b_2 = \frac{1}{1008}, \quad b_3 = -\frac{1}{1440}, \quad \dots \quad (2.10)$$

The displayed form follows either directly from a standard Barnes expansion or by inserting Stirling's series into the DLMF identity for  $\log G(z+1)$ . Nemes gives explicit remainder bounds and an exponentially improved form [11].

### 2.3 Normal-form principle

A translation  $x = z + \theta$  and a target shift  $Y \mapsto Y - c$  do not change the inversion problem, but they can remove entire slow blocks. The Gamma translation  $\theta = \frac{1}{2}$  changes the standard Stirling series into Equation (2.3), eliminating a logarithmic block at outer order zero and revealing an odd parity. For Barnes  $G$ , the natural recurrence already singles out  $z = x - 1$ ; the linear term  $\frac{1}{2}z$  is retained and generates the first inverse correction.

#### Practical normalization rule

Before reversing a power-logarithmic expansion, choose translations and constant target shifts to maximize the valuation of the remainder. A single removed low-order block can simplify every subsequent coefficient.

## 3 Exact Lambert normalization of a power-logarithmic leader

### 3.1 The basic leader

Let

$$F_0(x) = ax^p(\log x - \beta), \quad a \neq 0, \quad p \neq 0. \quad (3.1)$$

To solve  $F_0(X) = Y$ , set

$$w = W_k\left(\frac{pY}{a}e^{-p\beta}\right), \quad X = \exp\left(\beta + \frac{w}{p}\right) = \left(\frac{pY}{aw}\right)^{1/p}. \quad (3.2)$$

Then

$$p(\log X - \beta) = w, \quad aX^p \frac{w}{p} = Y. \quad (3.3)$$

On the positive real axis with  $a, p, Y > 0$ , the relevant branch is  $W_0$ .

**Proposition 3.1** (Exact dominant inversion). *The value  $X$  in Equation (3.2) solves  $F_0(X) = Y$  exactly on every compatible choice of logarithm, root, and Lambert branch.*

*Proof.* Put  $u = p(\log X - \beta)$ . Then

$$F_0(X) = \frac{a}{p} e^{p\beta} u e^u.$$

Thus  $F_0(X) = Y$  is equivalent to  $u e^u = (pY/a) e^{-p\beta}$ , so  $u = W_k((pY/a) e^{-p\beta})$ . Exponentiating  $\log X = \beta + u/p$  proves the claim.  $\square$

The derivative at the leading inverse is

$$F'_0(X) = aX^{p-1}(w+1). \quad (3.4)$$

The ubiquitous denominator  $w+1$  in inverse coefficients is therefore not accidental: it is the normalized Jacobian of the dominant map.

### 3.2 A general logarithmic power

The same exact reduction works for

$$F_{0,\rho}(x) = ax^p(\log x - \beta)^\rho, \quad \rho \neq 0. \quad (3.5)$$

Choose compatible branches and define

$$w = W_k\left(\frac{p}{\rho} \left(\frac{Y}{ae^{p\beta}}\right)^{1/\rho}\right), \quad X = \exp\left(\beta + \frac{\rho}{p}w\right). \quad (3.6)$$

Then  $F_{0,\rho}(X) = Y$ . The normalized linear coefficient in an additive correction is proportional to

$$p + \frac{\rho}{\log X - \beta} = p \left(1 + \frac{1}{w}\right), \quad (3.7)$$

so the same triangular inversion mechanism applies away from  $w = -1$ . The remainder of the article develops explicit universal formulas for the case  $\rho = 1$ , which includes both special functions of interest.

## 4 A formal reversion calculus in the Lambert coordinate

### 4.1 The transseries class

Consider a forward expansion

$$F(x) \sim ax^p(\log x - \beta) + \sum_{m \geq 1} x^{p-m} P_m(\log x), \quad (4.1)$$

where the  $P_m$  may be polynomials, rational functions, or elements of a slow differential field containing iterated logarithms. Put

$$R_m(s) = \frac{P_m(s)}{a}, \quad s = \log X = \beta + \frac{w}{p}, \quad q = X^{-1}, \quad (4.2)$$

and seek

$$x = X + U(q, w), \quad U(q, w) = \sum_{n \geq 0} C_n(w) q^n. \quad (4.3)$$

The slow variable  $w = p(s - \beta)$  grows logarithmically while  $q$  is the primary small parameter. Formal series are taken in the complete  $q$ -adic ring

$$\mathcal{T} = \mathcal{C}(w)[[q]], \quad (4.4)$$

where  $\mathcal{C}(w)$  is closed under the derivatives and Taylor translations required below.



## 4.2 The normalized residual equation

Since  $x = X(1 + qU)$ , direct substitution into Equation (4.1) leads to the normalized equation

$$\begin{aligned} \mathcal{E}(q, U; w) = & \frac{(1 + qU)^p \left( \frac{w}{p} + \log(1 + qU) \right) - \frac{w}{p}}{q} \\ & + \sum_{m \geq 1} q^{m-1} (1 + qU)^{p-m} R_m(s + \log(1 + qU)) = 0. \end{aligned} \quad (4.5)$$

Indeed,

$$F(X + U) - Y = aX^{p-1} \mathcal{E}(q, U; w). \quad (4.6)$$

Expanding  $\mathcal{E}$  at  $q = 0$  gives

$$\mathcal{E}(q, U; w) = (w + 1)U + R_1(s) + O(q). \quad (4.7)$$

Thus every new coefficient enters linearly with multiplier  $w + 1$ .

**Theorem 4.1** (Lambert-normalized formal inverse). *Assume that  $w + 1$  is invertible in the slow coefficient field. Then Equation (4.5) has a unique solution  $U \in \mathcal{C}(w)[[q]]$ . Its coefficients are determined triangularly by*

$$C_n(w) = -\frac{1}{w + 1} [q^n] \mathcal{E}(q, U_{<n}; w), \quad U_{<n} = \sum_{j=0}^{n-1} C_j(w) q^j. \quad (4.8)$$

If  $P_1 = \dots = P_{m_0-1} = 0$ , then  $C_0 = \dots = C_{m_0-2} = 0$ .

*Proof.* Suppose  $C_0, \dots, C_{n-1}$  are known and insert  $U = U_{<n} + C_n q^n + O(q^{n+1})$  into Equation (4.5). The coefficient of  $q^n$  contributed by the first derivative in  $U$  is  $(w + 1)C_n$ . Every nonlinear occurrence of  $C_n q^n$  carries at least one additional factor of  $q$ , and every lower block in the second line of Equation (4.5) does the same. Hence no other term at order  $q^n$  contains  $C_n$ . Solving the resulting linear equation gives Equation (4.8). Induction proves existence and uniqueness. The valuation statement follows because the first nonzero forward block  $P_{m_0}$  enters at order  $q^{m_0-1}$ .  $\square$

## 4.3 Closed coefficient form

For an explicit partition formula, define

$$f_j(w) = [t^j] (1 + t)^p \left( \frac{w}{p} + \log(1 + t) \right), \quad (4.9)$$

$$\mathcal{R}_{m,j}(s) = [t^j] (1 + t)^{p-m} R_m(s + \log(1 + t)). \quad (4.10)$$

Then  $f_1 = w + 1$ , and

$$\mathcal{E}(q, U; w) = \sum_{j \geq 1} f_j(w) q^{j-1} U^j + \sum_{m \geq 1} \sum_{j \geq 0} \mathcal{R}_{m,j}(s) q^{m+j-1} U^j. \quad (4.11)$$

Consequently,

$$\begin{aligned} (w + 1)C_n = & - \sum_{j=2}^{n+1} f_j(w) [q^{n-j+1}] U_{<n}^j \\ & - \sum_{m=1}^{n+1} \sum_{j=0}^{n-m+1} \mathcal{R}_{m,j}(s) [q^{n-m-j+1}] U_{<n}^j. \end{aligned} \quad (4.12)$$

Only finitely many terms contribute at each  $n$ . Formula Equation (4.12) is convenient for symbolic implementation with Bell polynomials or ordinary truncated-series arithmetic.

#### 4.4 The first three universal coefficients

Write

$$d = w + 1, \quad r_m = R_m(s), \quad \dot{r}_m = \frac{dr_m}{ds}. \quad (4.13)$$

Define

$$\Lambda_2 = \frac{(p-1)w + 2p - 1}{2}, \quad (4.14)$$

$$\Lambda_3 = \frac{(p-1)(p-2)w + 3p^2 - 6p + 2}{6}, \quad (4.15)$$

$$B_1 = (p-1)r_1 + \dot{r}_1, \quad (4.16)$$

$$B_2 = \binom{p-1}{2}r_1 + \left(p - \frac{3}{2}\right)\dot{r}_1 + \frac{1}{2}\ddot{r}_1, \quad (4.17)$$

$$D_2 = (p-2)r_2 + \dot{r}_2. \quad (4.18)$$

Then

$$C_0 = -\frac{r_1}{d}, \quad (4.19)$$

$$C_1 = -\frac{\Lambda_2 C_0^2 + B_1 C_0 + r_2}{d}, \quad (4.20)$$

$$C_2 = -\frac{2\Lambda_2 C_0 C_1 + \Lambda_3 C_0^3 + B_1 C_1 + B_2 C_0^2 + D_2 C_0 + r_3}{d}. \quad (4.21)$$

These formulas already recover the first three Barnes coefficients and explain the common denominator structure.

#### 4.5 Near-identity reduction and Lagrange–Buermann inversion

There is a second, coordinate-free derivation. Let

$$\Psi(x) = F_0^{(-1)}(F(x)) = x + H(x). \quad (4.22)$$

If  $F(x) = Y$  and  $X = F_0^{(-1)}(Y)$ , then  $\Psi(x) = X$ , so

$$x = \Psi^{(-1)}(X). \quad (4.23)$$

Moreover,

$$H(x) \sim \frac{F(x) - F_0(x)}{F_0'(x)} + \dots, \quad (4.24)$$

which belongs to the slow  $q$ -adic algebra. Differentiation with respect to  $X$  raises the primary  $q$ -valuation because

$$\frac{d}{dX}(q^n A(w)) = q^{n+1}(pA'(w) - nA(w)). \quad (4.25)$$

The usual Lagrange–Buermann formula is therefore summable blockwise.

**Theorem 4.2** (Lagrange representation at infinity). *Under the hypotheses of Theorem 4.1,*

$$F^{(-1)}(Y) = X + \sum_{n \geq 1} \frac{(-1)^n}{n!} \left( \frac{d}{dX} \right)^{n-1} H(X)^n, \quad H = F_0^{(-1)} \circ F - \text{id}. \quad (4.26)$$

*At every fixed power of  $q$ , only finitely many summands contribute.*

*Proof.* Apply the formal Lagrange–Buermann formula to the near-identity map  $X \mapsto X + H(X)$ . By Equation (4.25), the  $n$ -th summand has primary valuation at least  $n - 1$ . Hence the family is summable in the  $q$ -adic topology, and the resulting series is the unique inverse tangent to the identity.  $\square$

The direct residual recurrence Equation (4.8) is usually more efficient, while Equation (4.26) clarifies the relation to classical series reversion and provides an independent check.

## 4.6 Newton acceleration in the formal ring

Starting with any truncation  $U^{[0]}$ , define

$$U^{[j+1]} = U^{[j]} - \frac{\mathcal{E}(q, U^{[j]}; w)}{\partial_U \mathcal{E}(q, U^{[j]}; w)}. \quad (4.27)$$

Since  $\partial_U \mathcal{E} = (w + 1) + O(q)$  is invertible, Newton iteration approximately doubles the number of correct  $q$ -coefficients at each step. This is preferable to coefficient-by-coefficient recursion at very high order.

## 5 From formal series to asymptotic and exponential sectors

### 5.1 Residual certification

The normalized equation gives a particularly simple error test. Suppose a truncation  $U_N$  satisfies

$$\mathcal{E}(q, U_N; w) = O(q^{N+1} M(w)) \quad (5.1)$$

for a slow coefficient  $M$ . Then

$$F(X + U_N) - Y = O\left(aX^{p-N-2} M(w)\right). \quad (5.2)$$

If  $F$  is real, eventually increasing, and

$$F'(x) \geq cX^{p-1}|w + 1| \quad (5.3)$$

through the interval joining the exact and approximate roots, the mean value theorem gives

$$F^{(-1)}(Y) - (X + U_N) = O\left(\frac{q^{N+1} M(w)}{w + 1}\right). \quad (5.4)$$

This turns symbolic cancellation of the residual into an analytic error estimate once a differentiated forward asymptotic expansion is available.

For numerical work one can use the a posteriori bound

$$|x - \widehat{x}| \leq \frac{|F(\widehat{x}) - Y|}{\inf_I |F'|}, \quad (5.5)$$

where  $I$  joins  $x$  and  $\widehat{x}$ . A single ordinary Newton step then typically squares the already small transseries error.

## 5.2 Transport of genuinely exponential sectors

Suppose an exponentially improved forward transseries has the schematic form

$$F(x; \sigma) = F_{\text{alg}}(x) + \sum_{m \neq 0} \sigma^m R_m(x), \quad (5.6)$$

where  $R_m$  contains an exponential monomial such as  $\exp[-m \cdot \omega x^\nu]$ . Let  $x_{\text{alg}} = F_{\text{alg}}^{(-1)}(Y)$  and seek

$$x = x_{\text{alg}} + \sum_{m \neq 0} \sigma^m \Delta_m(x_{\text{alg}}). \quad (5.7)$$

The first sector is universal:

$$\Delta_{e_j}(x_{\text{alg}}) = -\frac{R_{e_j}(x_{\text{alg}})}{F'_{\text{alg}}(x_{\text{alg}})}. \quad (5.8)$$

Higher sectors are obtained recursively by expanding  $F(x_{\text{alg}} + \Delta; \sigma) - Y$  and extracting one exponential weight at a time. They contain derivatives of lower sectors and the usual Bell-polynomial couplings.

**Theorem 5.1** (Transseries implicit inversion). *In a well-based transseries field, if  $F'_{\text{alg}}(x_{\text{alg}})$  is invertible, the sector equations generated by Equations (5.6) and (5.7) are triangular in exponential weight. They have a unique solution tangent to  $x_{\text{alg}}$ , and their first equation is Equation (5.8).*

*Proof.* Expand each forward sector at  $x_{\text{alg}} + \Delta$ . At a fixed nonzero exponential weight  $m$ , the only term containing the new unknown  $\Delta_m$  linearly is  $F'_{\text{alg}}(x_{\text{alg}})\Delta_m$ . Every other contribution is a product or derivative of lower-weight sectors. Well-basedness ensures that only finitely many decompositions contribute to a prescribed weight. Division by the invertible derivative gives the unique recursion.  $\square$

The Bernoulli coefficients in both Stirling series grow factorially. Least-term truncation occurs at an index proportional to  $2\pi X$  or  $2\pi Z$ ; hence the remaining forward error has exponential order  $\exp[-2\pi X + O(\log X)]$  or  $\exp[-2\pi Z + O(\log Z)]$ . The corresponding inverse scales are, schematically,

$$\delta x_\Gamma = \frac{\exp[-2\pi X + O(\log X)]}{\log X}, \quad (5.9)$$

$$\delta x_G = \frac{\exp[-2\pi Z + O(\log Z)]}{Z \log Z}. \quad (5.10)$$

Precise Stokes multipliers are inherited from the exponentially improved forward expansions. For Barnes  $G$ , such expansions and Berry smoothing are developed by Nemes [11]; analogous exponential improvement for Stirling's series is classical [6, 13].

## 6 The increasing inverse Gamma function

### 6.1 Lambert coordinates

Let  $y \rightarrow +\infty$ , set

$$Y = \log y, \quad T = Y - c_\Gamma = \log\left(\frac{y}{\sqrt{2\pi}}\right), \quad (6.1)$$

and define

$$w = W_0(T/e), \quad X = e^{w+1} = \frac{T}{w}, \quad s = w + 1 = \log X, \quad q = X^{-1}. \quad (6.2)$$

Then

$$X(\log X - 1) = T. \quad (6.3)$$

Writing

$$\Gamma_+^{(-1)}(y) = \frac{1}{2} + X + U_\Gamma(q, s) \quad (6.4)$$

and using Equation (2.3) gives

$$\begin{aligned} \mathcal{E}_\Gamma(q, U; s) &= \frac{(1 + qU)(s - 1 + \log(1 + qU)) - (s - 1)}{q} \\ &+ \sum_{k \geq 1} a_k q^{2k-1} (1 + qU)^{-(2k-1)} = 0. \end{aligned} \quad (6.5)$$

Its linear coefficient is  $s$ .

## 6.2 Parity and all-orders recurrence

**Proposition 6.1** (Odd inverse-power parity). *The unique formal solution of Equation (6.5) satisfies*

$$U_\Gamma(-q, s) = -U_\Gamma(q, s). \quad (6.6)$$

Consequently,

$$U_\Gamma(q, s) = \sum_{n \geq 1} A_{2n-1}(s) q^{2n-1}. \quad (6.7)$$

*Proof.* A direct check gives  $\mathcal{E}_\Gamma(-q, -U; s) = -\mathcal{E}_\Gamma(q, U; s)$ . Therefore, if  $U(q, s)$  is a solution, so is  $-U(-q, s)$ . Uniqueness from Theorem 4.1 makes them equal.  $\square$

The coefficient recursion is

$$A_n(s) = -\frac{1}{s} [q^n] \mathcal{E}_\Gamma \left( q, \sum_{j=0}^{n-1} A_j(s) q^j; s \right), \quad A_{2j} = 0. \quad (6.8)$$

At large  $s$ , the direct contribution of the next Stirling coefficient dominates:

$$A_{2n-1}(s) = -\frac{a_n}{s} + \mathcal{O}(s^{-2}). \quad (6.9)$$

Because  $a_n$  has factorial growth, the outer Gamma inverse series is an asymptotic, generally divergent, transseries.

## 6.3 Explicit expansion

**Theorem 6.2** (Right inverse Gamma through five correction sectors). *With  $T, w, X, s$  as in Equations (6.1) and (6.2),*

$$\begin{aligned} \Gamma_+^{(-1)}(y) &= \frac{1}{2} + X + \frac{1}{24sX} - \frac{14s^2 + 10s + 5}{5760s^3X^3} \\ &+ \frac{2232s^4 + 1176s^3 + 714s^2 + 350s + 105}{2903040s^5X^5} \\ &- \frac{822960s^6 + 292536s^5 + 136956s^4 + 68040s^3 + 32970s^2 + 12250s + 2625}{1393459200s^7X^7} \\ &+ \frac{309052800s^8 + 77920128s^7 + 26024592s^6 + 10019592s^5 + 4516028s^4}{367873228800s^9X^9} \\ &+ \frac{2023560s^3 + 812350s^2 + 242550s + 40425}{367873228800s^9X^9} + \mathcal{O}\left(\frac{1}{sX^{11}}\right). \end{aligned} \quad (6.10)$$

The expansion is Poincare-asymptotic on the increasing real branch. The same coefficient functions apply on compatible complex branches after replacing  $W_0$  and the logarithms by the selected branches.

*Proof.* Insert  $U = \sum A_{2n-1} q^{2n-1}$  into Equation (6.5). Applying Equation (6.8) successively gives

$$\begin{aligned} A_1 &= \frac{1}{24s}, \\ A_3 &= -\frac{14s^2 + 10s + 5}{5760s^3}, \\ A_5 &= \frac{2232s^4 + 1176s^3 + 714s^2 + 350s + 105}{2903040s^5}, \\ A_7 &= -\frac{822960s^6 + 292536s^5 + 136956s^4 + 68040s^3 + 32970s^2 + 12250s + 2625}{1393459200s^7}, \\ A_9 &= \frac{309052800s^8 + 77920128s^7 + 26024592s^6 + 10019592s^5 + 4516028s^4}{367873228800s^9} \\ &\quad + \frac{2023560s^3 + 812350s^2 + 242550s + 40425}{367873228800s^9}. \end{aligned}$$

Substitution cancels  $\mathcal{E}_\Gamma$  through order  $q^9$ ; the next possible term has order  $q^{11}$ . The differentiated Stirling remainder and Equation (5.4) promote the formal identity to the stated real asymptotic expansion.  $\square$

#### 6.4 Derivative and residual checks

If  $x = \Gamma_+^{(-1)}(y)$ , then

$$\frac{dx}{dY} = \frac{1}{\psi(x)}, \quad \frac{dx}{dy} = \frac{1}{y\psi(x)}, \quad (6.11)$$

where  $\psi$  is the digamma function. Differentiating a transseries truncation and comparing with Equation (6.11) is an independent coefficient check. For numerical refinement, use

$$x_{\text{new}} = x - \frac{\log \Gamma(x) - Y}{\psi(x)}. \quad (6.12)$$

### 7 Pole branches of the inverse Gamma function

The Lambert sector describes the branch tending to  $+\infty$ . Every pole of  $\Gamma$  gives a different, locally convergent inverse expansion. Let

$$x = -n + \varepsilon, \quad n = 0, 1, 2, \dots, \quad r = \frac{1}{y}. \quad (7.1)$$

The reciprocal Gamma function is entire and has a simple zero at  $-n$ :

$$\frac{1}{\Gamma(-n + \varepsilon)} = (-1)^n n! \varepsilon \Phi_n(\varepsilon), \quad (7.2)$$

$$\Phi_n(\varepsilon) = \prod_{k=1}^n \left(1 - \frac{\varepsilon}{k}\right) \frac{1}{\Gamma(1 + \varepsilon)}, \quad \Phi_n(0) = 1. \quad (7.3)$$

Thus  $r = (-1)^n n! \varepsilon \Phi_n(\varepsilon)$ , and ordinary Lagrange inversion gives a convergent power series.

**Theorem 7.1** (All pole branches). *For sufficiently small  $|r|$ , the inverse branch tending to  $-n$  is*

$$\Gamma_{[-n]}^{(-1)}(y) = -n + \sum_{m \geq 1} \frac{t^m}{m} [\varepsilon^{m-1}] \Phi_n(\varepsilon)^{-m}, \quad t = \frac{(-1)^n}{n!y}. \quad (7.4)$$

If  $\psi_n = \psi(n+1) = H_n - \gamma$ , then

$$\Gamma_{[-n]}^{(-1)}(y) = -n + t + \psi_n t^2 + \frac{3\psi_n^2 + H_n^{(2)} + \zeta(2)}{2} t^3 + O(t^4). \quad (7.5)$$

For positive  $y$ , the leading displacement from the pole has sign  $(-1)^n$ .

*Proof.* Equation (7.2) follows from

$$\Gamma(-n + \varepsilon) = \frac{\Gamma(1 + \varepsilon)}{\varepsilon(\varepsilon - 1) \cdots (\varepsilon - n)}.$$

Apply the classical Lagrange formula to  $t = \varepsilon \Phi_n(\varepsilon)$ . Expanding

$$\log \Phi_n(\varepsilon) = -\psi_n \varepsilon - \frac{H_n^{(2)} + \zeta(2)}{2} \varepsilon^2 + \frac{\zeta(3) - H_n^{(3)}}{3} \varepsilon^3 + \cdots$$

gives the displayed coefficients. □

## 7.1 The branch tending to zero

For  $n = 0$ ,  $t = r = 1/y$ . Expanding further yields

$$\begin{aligned} \Gamma_{-}^{(-1)}(y) &= r - \gamma r^2 + \frac{18\gamma^2 + \pi^2}{12} r^3 - \frac{8\gamma^3 + \pi^2\gamma + \zeta(3)}{3} r^4 \\ &\quad + \frac{7500\gamma^4 + 1500\pi^2\gamma^2 + 2400\gamma\zeta(3) + 29\pi^4}{1440} r^5 \\ &\quad - \frac{1296\gamma^5 + 360\pi^2\gamma^3 + 720\gamma^2\zeta(3) + 17\pi^4\gamma}{120} r^6 \\ &\quad - \frac{24\zeta(5) + 20\pi^2\zeta(3)}{120} r^6 + O(r^7). \end{aligned} \quad (7.6)$$

Unlike Equation (6.10), this is a genuinely convergent local series because it inverts an analytic map with nonzero derivative at the origin.

## 8 The eventual positive inverse of Barnes G

### 8.1 Lambert coordinates and normalized equation

Let  $y \rightarrow +\infty$ , put

$$Y = \log y, \quad w = W_0(4Ye^{-3}), \quad Z = e^{(w+3)/2} = 2\sqrt{\frac{Y}{w}}, \quad (8.1)$$

and define

$$D = w + 1, \quad s = \log Z = \frac{D + 2}{2}, \quad q = Z^{-1}. \quad (8.2)$$

Then

$$\frac{1}{2}Z^2 \log Z - \frac{3}{4}Z^2 = Y. \quad (8.3)$$

Write

$$G_+^{(-1)}(y) = 1 + Z + U_G(q, D), \quad U_G = \sum_{n \geq 0} C_n(D) q^n. \quad (8.4)$$

Dividing the residual by  $\frac{1}{2}Z$  gives

$$\begin{aligned} \mathcal{E}_G(q, U; D) = & \frac{(1 + qU)^2 \left( \frac{w}{2} + \log(1 + qU) \right) - \frac{w}{2}}{q} + \lambda(1 + qU) \\ & + q \left[ -\frac{1}{6} (s + \log(1 + qU)) + 2\kappa \right] \\ & + \sum_{k \geq 1} 2b_k q^{2k+1} (1 + qU)^{-2k} = 0. \end{aligned} \quad (8.5)$$

The coefficient of  $U$  at  $q = 0$  is  $D$ , and the linear Barnes term immediately gives  $C_0 = -\lambda/D$ .

## 8.2 Explicit coefficient functions

**Theorem 8.1** (Barnes inverse through  $Z^{-3}$ ). *Let  $Y, w, Z, D$  be given by [Equations \(8.1\) and \(8.2\)](#), and let  $\lambda = \log(2\pi)$ ,  $\alpha = \log A$ . Then*

$$G_+^{(-1)}(y) = 1 + Z + C_0 + \frac{C_1}{Z} + \frac{C_2}{Z^2} + \frac{C_3}{Z^3} + O(Z^{-4}), \quad (8.6)$$

where

$$C_0 = -\frac{\lambda}{D}, \quad (8.7)$$

$$C_1 = \frac{1}{12} + \frac{2\alpha}{D} + \frac{\lambda^2}{2D^2} - \frac{\lambda^2}{D^3}, \quad (8.8)$$

$$C_2 = \frac{4\lambda\alpha}{D^3} + \frac{4\lambda^3}{3D^4} - \frac{2\lambda^3}{D^5}, \quad (8.9)$$

$$\begin{aligned} C_3 = & -\frac{1}{288} + \frac{11/720 - \alpha/6}{D} - \frac{2\alpha^2 + \lambda^2/24}{D^2} \\ & - \frac{4\alpha^2 + \alpha\lambda^2 + \lambda^2/12}{D^3} - \frac{2\alpha\lambda^2 + \lambda^4/8}{D^4} \\ & + \frac{12\alpha\lambda^2 - \lambda^4/6}{D^5} + \frac{25\lambda^4}{6D^6} - \frac{5\lambda^4}{D^7}. \end{aligned} \quad (8.10)$$

*Proof.* Apply the recurrence [Equation \(4.8\)](#) to [Equation \(8.5\)](#). The first equation is  $DC_0 + \lambda = 0$ . At each subsequent order, all nonlinear terms contain only previously determined coefficients. Exact simplification with  $\kappa = \frac{1}{12} - \alpha$  gives [Equations \(8.8\) to \(8.10\)](#). Substitution into [Equation \(8.5\)](#) cancels its coefficients through order  $q^3$ , and residual promotion gives the real asymptotic statement.  $\square$

The next coefficient, useful for numerical work, remains reasonably compact:

$$\begin{aligned} C_4 = & \frac{11\lambda}{360D^2} + \frac{11\lambda}{360D^3} - \frac{\lambda(72\alpha^2 + \lambda^2)}{9D^4} \\ & - \frac{\lambda(72\alpha^2 + 16\alpha\lambda^2 + \lambda^2)}{3D^5} - \frac{4\lambda^3(50\alpha + 3\lambda^2)}{15D^6} \\ & + \frac{4\lambda^3(30\alpha - \lambda^2)}{3D^7} + \frac{14\lambda^5}{D^8} - \frac{14\lambda^5}{D^9}. \end{aligned} \quad (8.11)$$



Thus Equation (8.6) can be extended by  $C_4/Z^4$  with remainder  $O(Z^{-5})$ , provided the forward expansion is retained to the corresponding order.

### 8.3 A compact hand recurrence

For low orders it is useful to avoid re-expanding the full normalized equation. Let

$$a = \frac{\lambda}{2}, \quad b = -\frac{1}{12}, \quad c = \kappa, \quad d_1 = -\frac{1}{240}, \quad d = s - 1 = \frac{D}{2}. \quad (8.12)$$

Writing  $U = \sum c_n Z^{-n}$ , one obtains

$$c_0 = -\frac{a}{d}, \quad (8.13)$$

$$c_1 = -\frac{ac_0 + bs + c + (s/2)c_0^2}{d}, \quad (8.14)$$

$$c_2 = -\frac{ac_1 + bc_0 + c_0^3/6 + sc_0c_1}{d}, \quad (8.15)$$

$$c_3 = -\frac{ac_2 - (b/2)c_0^2 + bc_1 + d_1 - c_0^4/24 + (c_0^2c_1)/2 + sc_0c_2 + (s/2)c_1^2}{d}. \quad (8.16)$$

These are algebraically equivalent to Equations (8.7) to (8.10).

### 8.4 Derivative check and Newton refinement

Let

$$\Psi_G(x) = \frac{d}{dx} \log G(x). \quad (8.17)$$

Then

$$\frac{d}{dY} G_+^{(-1)}(e^Y) = \frac{1}{\Psi_G(G_+^{(-1)}(e^Y))}. \quad (8.18)$$

A numerical implementation can refine any transseries truncation by

$$x_{\text{new}} = x - \frac{\log G(x) - Y}{\Psi_G(x)}. \quad (8.19)$$

Since  $\Psi_G(1 + Z) \sim Z \log Z$ , the asymptotic seed is already deep in the quadratic convergence basin for large  $Y$ .

## 9 Nested-logarithm re-expansions and sector ordering

### 9.1 Lambert $W$ at infinity

For  $\xi \rightarrow \infty$ , put

$$L = \log \xi, \quad \ell = \log L. \quad (9.1)$$

The standard expansion of  $W_0(\xi)$  is a polynomial-logarithmic series in  $L^{-1}$  with coefficients in  $\mathbb{Q}[\ell]$  [4, 5]. Re-expanding the exact Lambert normalizers therefore gives conventional transseries in nested logarithms.

## 9.2 Gamma nested-log sector

For Gamma take  $\xi = T/e$ , so

$$L = \log(T/e), \quad \ell = \log L. \quad (9.2)$$

Then

$$X = \frac{T}{L} \left[ 1 + \frac{\ell}{L} + \frac{\ell(\ell-1)}{L^2} + \frac{\ell(\ell-2)(2\ell-1)}{2L^3} + \frac{\ell(6\ell^3 - 26\ell^2 + 27\ell - 6)}{6L^4} + O\left(\frac{\ell^5}{L^5}\right) \right]. \quad (9.3)$$

Combining this with Equation (6.10) gives a fully elementary nested-log transseries.

### Sector ordering matters

For every fixed  $N$ , the omitted term  $X/L^N$  still tends to infinity. Therefore the additive shift  $\frac{1}{2}$  is smaller than *every fixed* number of nested-log corrections to  $X$ . Likewise, the first Stirling inverse sector  $1/(24sX)$  lies beyond the constant sector. A correctly ordered answer either retains  $W$  exactly or treats the completed Lambert subseries as one block before appending  $\frac{1}{2}$  and the powers of  $X^{-1}$ .

The asymptotic hierarchy is

$$X \gg \frac{X}{L} \gg \frac{X}{L^2} \gg \cdots \gg 1 \gg \frac{1}{X \log X} \gg \frac{1}{X^3 \log X} \gg \cdots. \quad (9.4)$$

## 9.3 Barnes nested-log sector

For Barnes take

$$L = \log(4Ye^{-3}), \quad \ell = \log L. \quad (9.5)$$

Then

$$Z = \frac{2\sqrt{Y}}{\sqrt{L}} \left[ 1 + \frac{\ell}{2L} + \frac{\ell(3\ell-4)}{8L^2} + \frac{\ell(5\ell^2-16\ell+8)}{16L^3} + \frac{\ell(105\ell^3-568\ell^2+720\ell-192)}{384L^4} + O\left(\frac{\ell^5}{L^5}\right) \right]. \quad (9.6)$$

The first slow denominator has expansion

$$\frac{1}{w+1} = \frac{1}{L} + \frac{\ell-1}{L^2} + \frac{\ell^2-3\ell+1}{L^3} + \frac{\ell^3 - \frac{11}{2}\ell^2 + 6\ell - 1}{L^4} + O\left(\frac{\ell^4}{L^5}\right). \quad (9.7)$$

Thus, after the completed Lambert block, the Barnes hierarchy begins

$$Z \quad (\text{with all nested logs}), \quad 1, \quad -\frac{\lambda}{w+1}, \quad \frac{C_1}{Z}, \quad \frac{C_2}{Z^2}, \dots \quad (9.8)$$

Again, retaining  $W$  exactly is both cleaner and numerically superior.

## 9.4 Complex branches

Let  $Y_j = \text{Log } y + 2\pi i j$ . For Gamma, define

$$T_j = Y_j - c_\Gamma, \quad w_{j,k} = W_k(T_j/e), \quad X_{j,k} = T_j/w_{j,k}. \quad (9.9)$$

For Barnes, define

$$w_{j,k} = W_k(4Y_j e^{-3}), \quad Z_{j,k} = 2\sqrt{Y_j/w_{j,k}}, \quad (9.10)$$

with a compatible square-root branch. In sectors where the relevant Stirling expansions hold and the selected forward branch is univalent, the same rational coefficient functions give formal inverse sheets. This construction does not by itself resolve global branch cuts; it is a local asymptotic parametrization of the sheets at infinity. Global inverse Gamma branch geometry is discussed by Jeffrey and Watt [10].

## 10 Numerical verification

All values in this section were computed with 80 decimal digits. The “exact” inverse is the high-precision root of  $\log \Gamma(x) = Y$  or  $\log G(x) = Y$ . The tables report absolute error. They are intended as coefficient checks, not as optimized numerical algorithms.

### 10.1 Gamma

Let

$$\mathcal{A}_{2N-1}(Y) = \frac{1}{2} + X + \sum_{n=1}^N \frac{A_{2n-1}(s)}{X^{2n-1}}, \quad \mathcal{A}_{-1} = \frac{1}{2} + X. \quad (10.1)$$

Table 1: Absolute error  $|\mathcal{A}_m(Y) - \Gamma_+^{(-1)}(e^Y)|$ .

| $Y$  | $m = -1$                | $m = 1$                  | $m = 3$                  | $m = 5$                  | $m = 7$                  | $m = 9$                  |
|------|-------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 10   | $2.4064 \times 10^{-3}$ | $2.9453 \times 10^{-6}$  | $1.2976 \times 10^{-8}$  | $1.3290 \times 10^{-10}$ | $2.5991 \times 10^{-12}$ | $8.3584 \times 10^{-14}$ |
| 50   | $5.7806 \times 10^{-4}$ | $8.0628 \times 10^{-8}$  | $4.6010 \times 10^{-11}$ | $6.2767 \times 10^{-14}$ | $1.6224 \times 10^{-16}$ | $6.8367 \times 10^{-19}$ |
| 100  | $3.0464 \times 10^{-4}$ | $1.5313 \times 10^{-8}$  | $3.2682 \times 10^{-12}$ | $1.6759 \times 10^{-15}$ | $1.6235 \times 10^{-18}$ | $2.5579 \times 10^{-21}$ |
| 1000 | $3.4011 \times 10^{-5}$ | $4.4431 \times 10^{-11}$ | $2.6678 \times 10^{-16}$ | $3.8769 \times 10^{-21}$ | $1.0573 \times 10^{-25}$ | $4.6656 \times 10^{-30}$ |

The corresponding exact values begin

$$\begin{aligned} \Gamma_+^{(-1)}(e^{10}) &= 8.715310138772789660993299 \dots, \\ \Gamma_+^{(-1)}(e^{50}) &= 23.48931723598978590000318 \dots, \\ \Gamma_+^{(-1)}(e^{100}) &= 38.18456559880504784450257 \dots, \\ \Gamma_+^{(-1)}(e^{1000}) &= 226.5074413075941080947948 \dots \end{aligned}$$

### 10.2 Barnes $G$

Define

$$\mathcal{B}_N(Y) = 1 + Z + \sum_{n=0}^N \frac{C_n(D)}{Z^n}, \quad \mathcal{B}_{-1} = 1 + Z. \quad (10.2)$$

Table 2: Absolute error  $|\mathcal{B}_N(Y) - G_+^{(-1)}(e^Y)|$ .

| $Y$   | $N = -1$                | $N = 0$                 | $N = 1$                 | $N = 2$                 | $N = 3$                  | $N = 4$                  |
|-------|-------------------------|-------------------------|-------------------------|-------------------------|--------------------------|--------------------------|
| 10    | $9.3803 \times 10^{-1}$ | $5.5074 \times 10^{-2}$ | $9.5222 \times 10^{-3}$ | $5.4316 \times 10^{-4}$ | $1.3042 \times 10^{-4}$  | $3.7180 \times 10^{-5}$  |
| 50    | $6.3838 \times 10^{-1}$ | $3.1685 \times 10^{-2}$ | $1.2981 \times 10^{-3}$ | $5.1633 \times 10^{-5}$ | $7.7735 \times 10^{-6}$  | $1.8663 \times 10^{-8}$  |
| 100   | $5.5115 \times 10^{-1}$ | $2.2811 \times 10^{-2}$ | $5.1226 \times 10^{-4}$ | $2.4782 \times 10^{-5}$ | $1.5720 \times 10^{-6}$  | $4.4894 \times 10^{-8}$  |
| 1000  | $3.6598 \times 10^{-1}$ | $7.0937 \times 10^{-3}$ | $2.3648 \times 10^{-5}$ | $9.6114 \times 10^{-7}$ | $5.4021 \times 10^{-9}$  | $2.6721 \times 10^{-10}$ |
| 10000 | $2.6677 \times 10^{-1}$ | $2.1957 \times 10^{-3}$ | $1.2367 \times 10^{-6}$ | $3.1269 \times 10^{-8}$ | $9.7000 \times 10^{-12}$ | $8.4220 \times 10^{-13}$ |

The exact roots begin

$$\begin{aligned}
G_+^{(-1)}(e^{10}) &= 6.919323766771216637535535 \dots, \\
G_+^{(-1)}(e^{50}) &= 11.07407747566167840724942 \dots, \\
G_+^{(-1)}(e^{100}) &= 13.92651070973114185567169 \dots, \\
G_+^{(-1)}(e^{1000}) &= 32.55204340388853580099282 \dots, \\
G_+^{(-1)}(e^{10000}) &= 83.54283578073789743023524 \dots
\end{aligned}$$

At  $Y = 10$ , the Barnes scale is only moderately large and the asymptotic character is visible. By  $Y = 50$ , the  $C_4$ -truncation already gains about eight digits, and the improvement accelerates rapidly.

## 11 Algorithms and symbolic implementation

### 11.1 Coefficient-generation algorithm

For a forward expansion of the form [Equation \(4.1\)](#), the direct algorithm is:

1. choose the leading Lambert coordinate  $w, X, q$ ;
2. build the normalized residual  $\mathcal{E}(q, U; w)$ ;
3. initialize  $U = 0$ ;
4. at order  $n$ , append an unknown  $c_n q^n$ , extract  $[q^n] \mathcal{E}$ , solve the resulting linear equation, and simplify;
5. substitute the truncation back into  $\mathcal{E}$  to verify the residual order.

Only forward coefficients through outer order  $n + 1$  are needed for  $C_n$ . The method works over exact rational-function fields, so no floating-point recognition is necessary.

### 11.2 Generic SymPy skeleton

The following compact code implements [Equation \(4.8\)](#) for  $P_m(s)$  supplied as exact SymPy expressions. It is deliberately minimal; production code should cache truncated powers and use Newton series iteration at high order.

```

import sympy as sp

def lambert_inverse_coefficients(p, w, s, R, order):
    """Return C_0, ..., C_order for the normalized equation (4.5).

    R is a dict {m: R_m(s)}. Missing m are interpreted as zero.
    The relation s = beta + w/p is imposed by the caller.
    """
    q = sp.Symbol("q")

```

```

coeffs = []
U = sp.Integer(0)

for n in range(order + 1):
    cn = sp.Symbol(f'c{n}')
    trial = U + cn*q**n
    h = sp.log(1 + q*trial)

    E = ((1 + q*trial)**p * (w/p + h) - w/p) / q
    for m in range(1, order + 2):
        Rm = R.get(m, sp.Integer(0))
        if Rm != 0:
            shifted = Rm.subs(s, s + h)
            E += q**(m - 1) * (1 + q*trial)**(p - m) * shifted

    eqn = sp.series(E, q, 0, n + 1).remove0().expand().coeff(q, n)
    value = sp.solve(sp.Eq(eqn, 0), cn)[0]
    value = sp.factor(sp.cancel(value))
    coeffs.append(value)
    U += value*q**n

return coeffs

```

For Gamma, use  $p = 1$ ,  $w = s - 1$ , and  $R_{2k} = a_k$ . For Barnes, use  $p = 2$ ,  $w = 2s - 3$ ,

$$R_1 = \lambda, \quad R_2 = -\frac{s}{6} + 2\kappa, \quad R_{2k+2} = 2b_k. \quad (11.1)$$

The output reproduces every coefficient displayed in this article.

### 11.3 Independent verification protocol

A robust symbolic workflow performs three checks:

1. *Residual check*: substitute the truncation into the normalized equation and verify that all targeted powers of  $q$  vanish.
2. *Derivative check*: differentiate the truncation and compare with the exact inverse-derivative identity.
3. *Numerical check*: evaluate at several separated target sizes and compare with high-precision root finding.

Agreement of all three checks makes transcription or branch errors much less likely.

## 12 Extensions and structural consequences

### 12.1 More general slow coefficient fields

Nothing in [Theorem 4.1](#) requires  $P_m$  to be a polynomial in one logarithm. One may take a slow field generated by

$$\ell_1 = \log X, \quad \ell_{j+1} = \log \ell_j, \quad (12.1)$$

closed under the slow derivation

$$\delta = \frac{\partial}{\partial \ell_1} + \frac{1}{\ell_1} \frac{\partial}{\partial \ell_2} + \frac{1}{\ell_1 \ell_2} \frac{\partial}{\partial \ell_3} + \cdots. \quad (12.2)$$

Then  $dA/dX = q\delta A$ , so differentiation raises the primary valuation exactly as required. Hahn-field and transseries foundations for such constructions are developed by van der Hoeven [17].

## 12.2 Regular variation viewpoint

The leading maps  $x^p \log x$  are regularly varying with index  $p$ . General regular-variation theory guarantees asymptotic inverses and organizes slowly varying factors [1]. The Lambert normalization is a stronger, application-specific statement: it inverts the dominant slowly varying factor exactly and exposes an effective coefficient calculus for all lower blocks.

## 12.3 Multiple logarithmic factors

For leaders such as

$$ax^p (\log x)^\rho (\log \log x)^\sigma, \quad (12.3)$$

there is generally no one-step elementary Lambert reduction. A useful strategy is to invert the first two factors by Equation (3.6), treat  $(\log \log x)^\sigma$  as a slow multiplier, and solve the remaining slow equation by fixed-point iteration. Once a leading inverse  $X$  is known to relative error  $o(1)$ , the additive correction still lives in a  $q = X^{-1}$  transseries and the triangular residual method applies.

## 12.4 Parameter-dependent families

If  $a, p, \beta$ , or the coefficients  $P_m$  depend analytically on a parameter, all recurrences remain valid over the enlarged coefficient field. Uniformity requires keeping  $w + 1$  away from zero and controlling forward remainders uniformly. This is useful for generalized Gamma and multiple-Gamma families.

## 12.5 Coefficient growth and optimal truncation

The first contribution of a newly appearing forward coefficient  $P_m$  to the inverse is its negative divided by the normalized derivative  $w + 1$ . Consequently, factorial growth of forward Stirling coefficients is inherited by the inverse coefficients. For Gamma,

$$a_n \sim \frac{2(-1)^n (2n-2)!}{(2\pi)^{2n}}, \quad A_{2n-1}(s) \sim -\frac{a_n}{s}, \quad (12.4)$$

up to the conventional sign interpretation of Bernoulli asymptotics. This predicts least-term truncation near  $n \asymp \pi X$ . The analogous Barnes estimate follows from  $b_n$ , with least term near  $n \asymp \pi Z$ . These estimates agree with the exponential scales in Equations (5.9) and (5.10).

# 13 Conclusions

The large-value inverse Gamma and inverse Barnes  $G$  functions share a single structural mechanism. Their logarithms have a dominant power-logarithmic block, and that block is exactly invertible by Lambert  $W$ . In the adapted coordinate, the remaining inversion problem is a near-identity formal equation over a slow coefficient field. Its Jacobian is  $w + 1$ , its coefficient equations are triangular, and its analytic error is controlled directly by the substitution residual.

For Gamma, the half shift converts Stirling's series into an odd inverse power series and yields an inverse expansion containing only  $X^{-1}, X^{-3}, X^{-5}, \dots$  beyond the leading block. The separate pole branches are ordinary convergent inversions of the reciprocal Gamma function. For Barnes  $G$ , the same calculus begins with a nonzero slow correction  $-\log(2\pi)/(w + 1)$  and then produces explicit powers of  $Z^{-1}$ .

The resulting method is not tied to either special function. It applies to broad families with leaders  $ax^p (\log x - \beta)^\rho$ , admits iterated-log coefficient fields, supports Lagrange and Newton formulations, and

transports exponentially small sectors in a universal way. The exact Lambert representation should normally be retained as the primary answer; nested-log expansion is a secondary re-expression whose internal sector must be completed before smaller additive corrections are ordered.

## A Coefficient derivation details

### A.1 Expansion of the dominant normalized block

Let

$$\Phi(t; w) = (1+t)^p \left( \frac{w}{p} + \log(1+t) \right). \quad (\text{A.1})$$

Then

$$\Phi(t; w) - \Phi(0; w) = (w+1)t + \Lambda_2 t^2 + \Lambda_3 t^3 + \cdots, \quad (\text{A.2})$$

with  $\Lambda_2, \Lambda_3$  as in [Equations \(4.14\) and \(4.15\)](#). Since  $t = qU$ , division by  $q$  produces

$$(w+1)U + \Lambda_2 q U^2 + \Lambda_3 q^2 U^3 + \cdots. \quad (\text{A.3})$$

Similarly,

$$\begin{aligned} (1+t)^{p-1} R_1(s + \log(1+t)) \\ = r_1 + B_1 t + B_2 t^2 + O(t^3), \end{aligned} \quad (\text{A.4})$$

$$(1+t)^{p-2} R_2(s + \log(1+t)) = r_2 + D_2 t + O(t^2). \quad (\text{A.5})$$

Collecting powers of  $q$  in [Equations \(A.3\) to \(A.5\)](#) gives the universal coefficients [Equations \(4.19\) to \(4.21\)](#).

### A.2 Why the Barnes constants simplify

DLMF gives an asymptotic form involving  $z \log \Gamma(z+1)$  and Glaisher's constant. Substitution of Stirling's series yields a constant  $1/12 - \log A$ , which is  $\zeta'(-1)$ . The two contributions to  $z^{-2k}$  combine as

$$\frac{B_{2k+2}}{(2k+2)(2k+1)} + \frac{B_{2k+2}}{2k(2k+1)(2k+2)} = \frac{B_{2k+2}}{2k(2k+2)}. \quad (\text{A.6})$$

This proves [Equations \(2.7\) and \(2.9\)](#) from the DLMF form.

## B Further pole-branch coefficients

Let  $t = (-1)^n / (n!y)$ ,  $\psi_n = H_n - \gamma$ , and  $H_n^{(r)} = \sum_{k=1}^n k^{-r}$ . Continuing [Equation \(7.4\)](#) gives

$$\begin{aligned} \Gamma_{[-n]}^{(-1)}(y) = & -n + t + \psi_n t^2 + \frac{3\psi_n^2 + H_n^{(2)} + \zeta(2)}{2} t^3 \\ & + \frac{8\psi_n^3 + 6\psi_n(H_n^{(2)} + \zeta(2)) + H_n^{(3)} - \zeta(3)}{3} t^4 + O(t^5). \end{aligned} \quad (\text{B.1})$$

For  $n = 0$ , this reduces to the first four terms of [Equation \(7.6\)](#).

## C Notation and reproducibility checklist

| Symbol        | Meaning                                                           |
|---------------|-------------------------------------------------------------------|
| $f^{(-1)}$    | Compositional inverse, not reciprocal.                            |
| $Y$           | Logarithm of the large function value: $Y = \log y$ .             |
| $w$           | Lambert coordinate that exactly inverts the dominant block.       |
| $X, Z$        | Exact leading inverses for Gamma and Barnes $G$ , respectively.   |
| $q$           | Primary small parameter, $q = X^{-1}$ or $Z^{-1}$ .               |
| $s, D$        | Slow variables: $s = w + 1$ for Gamma and $D = w + 1$ for Barnes. |
| $A$           | Glaisher–Kinkelin constant; $\alpha = \log A$ .                   |
| $\mathcal{E}$ | Normalized substitution residual.                                 |

To reproduce the results:

1. generate Bernoulli or Bernoulli-polynomial coefficients exactly;
2. use Equation (6.5) or Equation (8.5);
3. apply Equation (4.8);
4. check cancellation in the normalized residual;
5. evaluate against high-precision roots of the logarithmic equations.

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